

A Discrete Dual Certificate for the Master Route Ratio in the A Priori TSP with Depot

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Abstract

Blauth, Neuwohner, Puhlmann, and Vygen conjectured that the master route ratio for the a priori traveling salesperson problem with a depot is

$$\frac{1}{1 - e^{-1/2}}.$$

A previously isolated continuous variational inequality gives the constant $1 - e^{-1/2}$, but by itself leaves a finite-to-continuum passage. This note writes the missing passage directly at the discrete level. For $h = p$, let $a_k = (1 - h)^{k-1}$. We prove that every nonnegative sequence $(f_k)_{k \geq 1}$ satisfying

$$f_{i+j} \leq f_i + f_j, \quad f_k + 2h \sum_{i=1}^{k-1} \min\{f_i, f_{k-i}\} \geq kh$$

has

$$h \sum_{k \geq 1} a_k f_k \geq 1 - e^{-1/2} - o(1) \quad (h \downarrow 0).$$

The proof is a discrete dual certificate. It combines a subadditivity certificate on the short range $kh \lesssim 1/2$ with a plateau-set domination inequality on the tail, followed by a layer-cake argument. The continuous certificate is recovered as the formal limit, while the discrete result is the statement needed for the small- p master-route-ratio optimization.

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1 Introduction

In the a priori TSP with independent activations, one is given a finite metric or semimetric space (V, c) , a depot $d \in V$ with activation probability $p(d) = 1$, and independent activation probabilities $p(v) \in (0, 1]$ for all nondepot customers. An a priori tour is fixed before the active set is known. When the active set is realized, inactive vertices are skipped.

A master-route solution chooses a set $S \subseteq V$ containing the depot, builds a TSP tour on S , and connects every vertex outside S to a nearest point of S by two parallel edges. Let $\text{MR}(S)$ denote the usual master-route upper bound on the expected realized cost and let OPT denote the optimal expected a priori tour cost. The master route ratio with depot is

$$\rho^* := \sup_{(V, c, p, d)} \min_{S \ni d} \frac{\text{MR}(S)}{\text{OPT}}.$$

Blauth, Neuwohner, Puhlmann, and Vygen conjectured that

$$\rho^* = \frac{1}{1 - e^{-1/2}}.$$

The lower bound is supplied by a clustered construction. The upper bound is reduced in [1] to a one-dimensional master-route optimization. In the equal-probability, small-activation regime the optimization has variables C_k satisfying

$$C_{i+j} \leq C_i + C_j, \tag{1}$$

$$C_k + 2p \sum_{i=1}^{k-1} \min\{C_i, C_{k-i}\} \geq kp^2, \tag{2}$$

and the objective is

$$\sum_{k \geq 1} (1 - p)^{k-1} C_k.$$

The formal scaling $x = kp$, $C_k \approx pF(kp)$ leads to the continuous variational inequality

$$F(x) + 2 \int_0^x \min\{F(y), F(x - y)\} dy \geq x,$$

whose sharp objective value is $1 - e^{-1/2}$. The purpose of this manuscript is to prove the corresponding lower bound directly for the discrete problem, with no compactness or unproved convergence assertion.

The main discrete theorem is the following.

Theorem 1.1 (Discrete certificate). *Let $h \in (0, 1)$ and set $a_k = (1 - h)^{k-1}$. Let $(f_k)_{k \geq 1}$ be a nonnegative sequence such that*

$$f_{i+j} \leq f_i + f_j \quad (i, j \geq 1), \quad (3)$$

$$f_k + 2h \sum_{i=1}^{k-1} \min\{f_i, f_{k-i}\} \geq kh \quad (k \geq 1). \quad (4)$$

Then

$$h \sum_{k \geq 1} a_k f_k \geq 1 - e^{-1/2} - o(1) \quad (h \downarrow 0),$$

where the $o(1)$ is uniform over all feasible sequences.

Indeed, if $C_k = hf_k$, then (1)–(2) become (3)–(4), and the objective becomes $h \sum_k a_k f_k$. Thus Theorem 1.1 is precisely the finite-to-continuum step in the small- p master-route optimization.

For completeness, Section 2 recalls the clustered lower bound. Sections 3–6 prove Theorem 1.1. Appendix A records the limiting continuous certificate; it is not needed for the discrete proof, but it explains the origin of the constants.

2 The clustered lower bound

We recall the standard construction giving

$$\rho^* \geq \frac{1}{1 - e^{-1/2}}.$$

Let n, m be positive integers. Consider a depot d and n clusters $V_1, \dots, V_n$, each containing m nondepot customers. Define a semimetric by

$$c(d, v) = 1, \quad c(v, w) = \begin{cases} 0, & v, w \text{ in the same cluster,} \\ 1, & v, w \text{ in different clusters.} \end{cases}$$

If a strict metric is desired, replace the within-cluster distance 0 by $\varepsilon > 0$ and let $\varepsilon \downarrow 0$. Each nondepot customer is active with probability

$$p = \frac{1}{2m}.$$

A cluster is active with probability

$$q_m = 1 - \left(1 - \frac{1}{2m}\right)^m \rightarrow 1 - e^{-1/2}.$$

Let $S \ni d$ be a master-route set, and let k be the number of nondepot clusters intersected by S . If $k = 0$, the master route $S = \{d\}$ has expected connection cost

$$2 \cdot nm \cdot \frac{1}{2m} = n.$$

If $k \geq 1$, the TSP tour on S has length at least $k + 1$, and the expected connection cost from the clusters not intersected by S is at least $n - k$. Let

$$P_{n,m} := 1 - \left(1 - \frac{1}{2m}\right)^{nm}$$

be the probability that at least one nondepot customer is active. Then

$$\text{MR}(S) \geq n - k + (k + 1)P_{n,m} \geq n - n(1 - P_{n,m}).$$

For fixed n , as $m \rightarrow \infty$ we have $P_{n,m} \rightarrow 1 - e^{-n/2}$. Hence, after taking $m \rightarrow \infty$ and then $n \rightarrow \infty$,

$$\min_{S \ni d} \text{MR}(S) \geq n - o(n).$$

On the other hand, the a priori tour that visits the clusters consecutively has realized cost at most $K + \mathbf{1}_{\{K \geq 1\}}$, where K is the number of active clusters. Therefore

$$\text{OPT} \leq \mathbb{E}K + 1 = nq_m + 1.$$

Consequently

$$\rho^* \geq \lim_{n \rightarrow \infty} \lim_{m \rightarrow \infty} \frac{n - o(n)}{nq_m + 1} = \frac{1}{1 - e^{-1/2}}.$$

3 Discrete notation and exponential sums

Throughout the discrete proof, put

$$r := e^{-1/2}, \quad \gamma := 1 - r, \quad \Lambda := 2 - 3r, \quad C := 4\Lambda = 8 - 12r.$$

Let

$$\beta := 1 - h, \quad a_k := \beta^{k-1}.$$

Choose the short-range cutoff

$$M = M(h) := \min\{m \geq 1 : a_m \leq r\}$$

and set

$$N = N(h) := 2M.$$

Then

$$Mh \rightarrow \frac{1}{2}, \quad Nh \rightarrow 1, \quad a_M \rightarrow r, \quad a_N \rightarrow r^2 \quad (h \downarrow 0).$$

More precisely, $a_M \leq r < a_{M-1}$ and $a_N = \beta a_M^2 \leq \beta r^2$.

Define the constants

$$P := \frac{7r - 4}{r^2}, \quad B := \frac{4(5r - 3)}{r^4}.$$

For all sufficiently small h , define a nonnegative discrete density by

$$\lambda_k := \begin{cases} 0, & k < N, \\ Pa_k - Ba_k^2, & k \geq N. \end{cases}$$

The nonnegativity follows because $a_k \leq a_N \leq \beta r^2$ for $k \geq N$ and

$$P - Br^2 = \frac{8 - 13r}{r^2} > 0.$$

Finally set

$$\Lambda_h := h \sum_{k \geq N} \lambda_k, \quad Q_h := h \sum_{k \geq N} kh \lambda_k.$$

Lemma 3.1 (Normalization limits). *As $h \downarrow 0$,*

$$\Lambda_h \rightarrow \Lambda = 2 - 3r, \quad Q_h \rightarrow \gamma = 1 - r.$$

Proof. Using the geometric sums

$$\sum_{k \geq N} a_k = \frac{a_N}{h}, \quad \sum_{k \geq N} a_k^2 = \frac{a_N^2}{h(2-h)},$$

we get

$$\Lambda_h = Pa_N - B \frac{a_N^2}{2-h} \rightarrow Pr^2 - \frac{B}{2} r^4.$$

Since $Pr^2 = 7r - 4$ and $Br^4 = 4(5r - 3)$, this limit is

$$(7r - 4) - 2(5r - 3) = 2 - 3r.$$

Similarly,

$$h^2 \sum_{k \geq N} ka_k = a_N \{Nh + \beta\},$$

and

$$h^2 \sum_{k \geq N} ka_k^2 = a_N^2 \left\{ \frac{Nh}{2-h} + \frac{\beta^2}{(2-h)^2} \right\}.$$

Therefore

$$Q_h \rightarrow 2Pr^2 - \frac{3}{4}Br^4 = 2(7r - 4) - 3(5r - 3) = 1 - r.$$

□

4 The short-range subadditivity certificate

The first ingredient controls the short range $1 \leq k < M$ using only subadditivity.

For $s > 0$ define

$$K(s) := \frac{1}{s^3} \int_0^s u^2 e^{-u} du,$$

and set $K(0) = 1/3$. The following identity is the continuum calibration behind the discrete construction.

Lemma 4.1 (Short-range calibration). *For every $0 \leq t \leq 1/2$,*

$$2 \int_t^{1/2} K(s) ds - tK(t) = e^{-t} - C.$$

Proof. Since $(t^3 K(t))' = t^2 e^{-t}$, we have

$$3K(t) + tK'(t) = e^{-t}.$$

Thus

$$\frac{d}{dt} \left(2 \int_t^{1/2} K(s) ds - tK(t) \right) = -e^{-t}.$$

At $t = 1/2$, the left-hand side is $-(1/2)K(1/2)$. A direct calculation gives

$$\frac{1}{2}K\left(\frac{1}{2}\right) = 8 - 13r.$$

Hence the endpoint value is

$$-(8 - 13r) = r - (8 - 12r) = e^{-1/2} - C.$$

Integration from the endpoint proves the identity. $\square$

For $2 \leq s \leq M - 1$, set

$$\kappa_s := h^2 K(sh).$$

For $1 \leq t \leq M - 1$, define

$$A_t := 2 \sum_{s=t+1}^{M-1} \kappa_s - (t-1)\kappa_t,$$

where the first sum is interpreted as 0 when $t = M - 1$. Finally let

$$D_h := \min_{1 \leq t \leq M-1} \left(a_t - \frac{A_t}{h} \right).$$

Lemma 4.2 (Discrete short-range domination). *For every nonnegative subadditive sequence (f_k) ,*

$$h \sum_{t=1}^{M-1} a_t f_t \geq D_h h \sum_{t=1}^{M-1} f_t.$$

Moreover

$$D_h = C + o(1) \quad (h \downarrow 0).$$

Proof. By subadditivity,

$$f_i + f_{s-i} - f_s \geq 0 \quad (1 \leq i \leq s-1).$$

Thus

$$0 \leq \sum_{s=2}^{M-1} \kappa_s \sum_{i=1}^{s-1} (f_i + f_{s-i} - f_s).$$

The coefficient of f_t in the last expression is exactly A_t . Therefore

$$\sum_{t=1}^{M-1} A_t f_t \geq 0.$$

By definition of D_h ,

$$h(a_t - D_h) \geq A_t \quad (1 \leq t \leq M-1).$$

Multiplying by f_t and summing gives

$$h \sum_{t=1}^{M-1} (a_t - D_h) f_t \geq \sum_{t=1}^{M-1} A_t f_t \geq 0,$$

which proves the domination inequality.

It remains to identify the limit of D_h . Uniformly for $1 \leq t \leq M-1$, with $x_t = th$,

$$\frac{A_t}{h} = 2h \sum_{s=t+1}^{M-1} K(sh) - (t-1)hK(th) = 2 \int_{x_t}^{1/2} K(s) ds - x_t K(x_t) + o(1).$$

Here we used the continuity of K on $[0, 1/2]$ and the fact that $Mh \rightarrow 1/2$. Also

$$a_t = (1-h)^{t-1} = e^{-x_t} + o(1)$$

uniformly for $1 \leq t \leq M-1$. Lemma 4.1 gives

$$a_t - \frac{A_t}{h} = C + o(1)$$

uniformly in t , and hence $D_h = C + o(1)$. □

5 The tail plateau certificate

The second ingredient is a discrete analogue of the plateau-set domination inequality.

Lemma 5.1 (Discrete compression). *Let $E \subseteq \mathbb{N}$, and put*

$$m = h \sum_{i \in E} a_i, \quad n = h \sum_{i \in E} a_i^2.$$

Then

$$n \geq \frac{m^2}{2-h}.$$

Proof. Write the selected values in decreasing order as $b_1, b_2, \dots$, omitting zero terms if the set is finite. Since the selected values are drawn from the geometric sequence $1, \beta, \beta^2, \dots$, we have $b_{j+d} \leq \beta^d b_j$ whenever both sides are present. Hence

$$\begin{aligned} \left(\sum_j b_j \right)^2 &= \sum_j b_j^2 + 2 \sum_{d \geq 1} \sum_j b_j b_{j+d} \\ &\leq \sum_j b_j^2 + 2 \sum_{d \geq 1} \beta^d \sum_j b_j^2 = \frac{2-h}{h} \sum_j b_j^2. \end{aligned}$$

Multiplying by h^2 gives

$$m^2 \leq (2-h)n. \quad \square$$

Lemma 5.2 (Discrete plateau-set domination). *For all sufficiently small h and every set $E \subseteq \{M, M+1, \dots\}$,*

$$h \sum_{i \in E} a_i \geq h \sum_{i \in E} \lambda_i + 2h^2 \sum_{u,v \in E} \lambda_{u+v}.$$

Proof. Let

$$E_1 := E \cap \{N, N+1, \dots\}.$$

Define

$$m = h \sum_{i \in E} a_i, \quad n = h \sum_{i \in E} a_i^2,$$

and

$$m_1 = h \sum_{i \in E_1} a_i, \quad n_1 = h \sum_{i \in E_1} a_i^2.$$

Because $E \subseteq \{M, M+1, \dots\}$,

$$0 \leq m \leq h \sum_{i \geq M} a_i = a_M \leq r,$$

and because $E_1 \subseteq \{N, N+1, \dots\}$,

$$0 \leq m_1 \leq h \sum_{i \geq N} a_i = a_N \leq \beta r^2 < r^2.$$

For $u, v \geq M$, we have $u+v \geq N$ and

$$a_{u+v} = \beta a_u a_v, \quad a_{u+v}^2 = \beta^2 a_u^2 a_v^2.$$

Thus the difference between the left-hand side and the right-hand side of the desired inequality is

$$G_h = m - Pm_1 + Bn_1 - 2P\beta m^2 + 2B\beta^2 n^2.$$

By Lemma 5.1, applied to E and E_1 ,

$$n \geq \frac{m^2}{2-h}, \quad n_1 \geq \frac{m_1^2}{2-h}.$$

Consequently

$$G_h \geq m - Pm_1 + \frac{B}{2-h} m_1^2 - 2P\beta m^2 + \frac{2B\beta^2}{(2-h)^2} m^4. \quad (5)$$

For fixed m , the right-hand side of (5) is decreasing in $m_1 \in [0, r^2]$ for all sufficiently small h , since

$$-P + \frac{2B}{2-h} m_1 \leq -P + \frac{2B}{2-h} r^2 < 0$$

when h is small; the limiting strict inequality is $-P + Br^2 < 0$, equivalent to $8 - 13r > 0$. Hence the worst case is obtained by replacing m_1 by $\min\{m, r^2\}$.

First suppose $0 \leq m \leq r^2$. Then (5) is bounded below by

$$H_{1,h}(m) := (1-P)m + \left(\frac{B}{2-h} - 2P\beta \right) m^2 + \frac{2B\beta^2}{(2-h)^2} m^4.$$

Let

$$A_h := \frac{B}{2-h} - 2P\beta.$$

For small h , $A_h < 0$. Therefore

$$H_{1,h}(m) \geq m \left((1 - P) + A_h r^2 \right).$$

At $h = 0$ the coefficient equals

$$(1 - P) + \left(\frac{B}{2} - 2P \right) r^2 = -\frac{(2r - 1)(7r^2 - r - 2)}{r^2} > 0.$$

Thus $H_{1,h}(m) \geq 0$ for all $0 \leq m \leq r^2$ and all sufficiently small h .

Now suppose $r^2 \leq m \leq r$. Then (5) is bounded below by

$$H_{2,h}(m) := m - Pr^2 + \frac{B}{2-h} r^4 - 2P\beta m^2 + \frac{2B\beta^2}{(2-h)^2} m^4.$$

At $m = r^2$ this agrees with $H_{1,h}(r^2)$, so it is nonnegative for small h . At $m = r$,

$$H_{2,h}(r) = h \left(2Pr^2 - \frac{1}{4}Br^4 \right) + O(h^2) = h(9r - 5) + O(h^2) > 0.$$

Moreover

$$H_{2,h}''(m) = -4P\beta + \frac{24B\beta^2}{(2-h)^2} m^2.$$

For $r^2 \leq m \leq r$, this is strictly negative for all small h , because at $h = 0$ it is at most

$$-4P + 6Br^2 = \frac{4(23r - 14)}{r^2} < 0.$$

Thus $H_{2,h}$ is concave on $[r^2, r]$ and has nonnegative endpoint values. Hence $H_{2,h}(m) \geq 0$ throughout this interval. Therefore $G_h \geq 0$, proving the lemma. $\square$

Remark 5.3. *The scalar signs used above are elementary. The numerical bound*

$$0.6065 < r = e^{-1/2} < 0.6066$$

implies

$$8 - 13r > 0, \quad 5r - 3 > 0, \quad 23r - 14 < 0, \quad 9r - 5 > 0, \quad 7r^2 - r - 2 < 0.$$

The only polynomial identity in the first case is

$$(1 - P) + \left(\frac{B}{2} - 2P \right) r^2 = -\frac{(2r - 1)(7r^2 - r - 2)}{r^2}.$$

This avoids the incorrect simplification $B/2 - 2P = 2(1 - 2r)/r^2$; that expression is not equal to $B/2 - 2P$ for the values of P, B above.

Lemma 5.4 (Discrete plateau functional domination). *For every nonnegative sequence $(f_k)_{k \geq 1}$ and all sufficiently small h ,*

$$h \sum_{k \geq M} a_k f_k \geq h \sum_{k \geq N} \lambda_k f_k + 2h^2 \sum_{u, v \geq M} \lambda_{u+v} \min\{f_u, f_v\}.$$

Proof. For $s \geq 0$, let

$$E_s := \{k \geq M : f_k > s\}.$$

Apply Lemma 5.2 to E_s and integrate over $s \in [0, \infty)$. Since all summands are nonnegative, Tonelli's theorem for sums gives

$$\begin{aligned} \int_0^\infty h \sum_{k \in E_s} a_k ds &= h \sum_{k \geq M} a_k f_k, \\ \int_0^\infty h \sum_{k \in E_s} \lambda_k ds &= h \sum_{k \geq N} \lambda_k f_k, \end{aligned}$$

and

$$\int_0^\infty 2h^2 \sum_{u,v \in E_s} \lambda_{u+v} ds = 2h^2 \sum_{u,v \geq M} \lambda_{u+v} \min\{f_u, f_v\}.$$

This proves the claim. $\square$

6 Proof of the discrete certificate

We now prove Theorem 1.1. Let (f_k) be feasible for (3)–(4). If $h \sum_k a_k f_k = \infty$, there is nothing to prove, so assume the objective is finite.

Set

$$\theta_h := \min \left\{ 1, \frac{D_h}{4\Lambda_h} \right\}.$$

By Lemmas 3.1 and 4.2,

$$\theta_h \rightarrow 1.$$

Moreover, since $\lambda_k \geq 0$, Lemma 5.4 remains true if every λ_k is replaced by $\theta_h \lambda_k$.

Multiply (4) by $h\theta_h \lambda_k$ and sum over $k \geq N$. Using the definition of Q_h gives

$$\theta_h Q_h \leq \theta_h h \sum_{k \geq N} \lambda_k f_k + 2\theta_h h^2 \sum_{k \geq N} \lambda_k \sum_{i=1}^{k-1} \min\{f_i, f_{k-i}\}. \quad (6)$$

Split the inner sum into the three ranges

$$1 \leq i < M, \quad M \leq i \leq k - M, \quad k - M < i \leq k - 1.$$

The middle range equals

$$2\theta_h h^2 \sum_{u,v \geq M} \lambda_{u+v} \min\{f_u, f_v\}.$$

Therefore Lemma 5.4, with $\theta_h \lambda$ in place of λ , implies that the first term in (6) plus the middle contribution is at most

$$h \sum_{k \geq M} a_k f_k.$$

For the two edge ranges, use $\min\{f_i, f_{k-i}\} \leq f_i$ on the left edge and $\min\{f_i, f_{k-i}\} \leq f_{k-i}$ on the right edge. The total edge contribution is at most

$$4\theta_h h^2 \left(\sum_{k \geq N} \lambda_k \right) \left(\sum_{i=1}^{M-1} f_i \right) = 4\theta_h \Lambda_h h \sum_{i=1}^{M-1} f_i.$$

By the definition of θ_h ,

$$4\theta_h\Lambda_h \leq D_h.$$

Thus the edge contribution is at most

$$D_h h \sum_{i=1}^{M-1} f_i,$$

which is at most

$$h \sum_{i=1}^{M-1} a_i f_i$$

by Lemma 4.2.

Combining these estimates in (6), we obtain

$$\theta_h Q_h \leq h \sum_{i=1}^{M-1} a_i f_i + h \sum_{k \geq M} a_k f_k = h \sum_{k \geq 1} a_k f_k.$$

Finally, $\theta_h \rightarrow 1$ and $Q_h \rightarrow 1 - r$ by Lemma 3.1. Hence

$$h \sum_{k \geq 1} a_k f_k \geq 1 - e^{-1/2} - o(1),$$

uniformly over all feasible sequences. This proves Theorem 1.1.

7 Consequence for the master route ratio

The clustered construction of Section 2 gives

$$\rho^* \geq \frac{1}{1 - e^{-1/2}}.$$

For the reverse inequality, use the master-route-ratio reduction of [1]: in the equal-probability, small- p normalization, every instance gives a feasible sequence (C_k) satisfying (1)–(2), and the reciprocal of the optimal value of the corresponding objective bounds the master-route ratio. Setting $h = p$ and $C_k = h f_k$, Theorem 1.1 gives the limiting lower bound

$$\sum_{k \geq 1} (1 - p)^{k-1} C_k \geq 1 - e^{-1/2} - o(1) \quad (p \downarrow 0).$$

Thus the small- p obstruction has value at least $1 - e^{-1/2}$. Together with the standard reduction in [1], this gives

$$\rho^* \leq \frac{1}{1 - e^{-1/2}}.$$

Combining the two inequalities yields

$$\rho^* = \frac{1}{1 - e^{-1/2}}.$$

A The limiting continuous certificate

This appendix records the continuous variational inequality obtained by formally sending $h \downarrow 0$ in the discrete certificate. It is included to show where the constants come from. The discrete proof above does not rely on compactness of near-minimizers.

Theorem A.1 (Continuous variational lemma). *Let $F : [0, \infty) \rightarrow [0, \infty)$ be locally integrable and subadditive:*

$$F(x + y) \leq F(x) + F(y) \quad (x, y \geq 0).$$

Assume moreover that, for every $x \geq 0$,

$$F(x) + 2 \int_0^x \min\{F(y), F(x - y)\} dy \geq x. \quad (7)$$

Then

$$\int_0^\infty e^{-x} F(x) dx \geq 1 - e^{-1/2}.$$

Equality is attained by $F_(x) = \min\{x, 1/2\}$.*

A.1 The short interval

The calibration Lemma 4.1 immediately gives the continuous short-interval domination.

Lemma A.2. *If F is nonnegative, locally integrable, and subadditive, then*

$$\int_0^{1/2} e^{-t} F(t) dt \geq C \int_0^{1/2} F(t) dt.$$

Proof. By subadditivity,

$$F(a) + F(b) - F(a + b) \geq 0.$$

Therefore

$$0 \leq \iint_{a, b \geq 0, a+b \leq 1/2} K(a + b) (F(a) + F(b) - F(a + b)) da db.$$

Changing variables $s = a + b$, the coefficient of $F(t)$ in this integral is

$$2 \int_t^{1/2} K(s) ds - tK(t).$$

By Lemma 4.1, the integral is

$$\int_0^{1/2} (e^{-t} - C) F(t) dt.$$

This is nonnegative, proving the claim. □

A.2 The plateau inequality

Define a density supported on $[1, \infty)$ by

$$\lambda(x) := \begin{cases} 0, & x < 1, \\ Pe^{-x} - Be^{-2x}, & x \geq 1. \end{cases}$$

Equivalently,

$$\lambda(x) = e^{-x}(a + b(1 - e^{-(x-1)})) \quad (x \geq 1),$$

where

$$a = e(8 - 13r), \quad b = 4e(5r - 3).$$

The normalization identities are

$$\int_1^\infty \lambda(x) dx = \Lambda = 2 - 3r, \quad \int_1^\infty x\lambda(x) dx = \gamma = 1 - r. \quad (8)$$

Lemma A.3 (Tail compression). *Let $\alpha \geq 0$, and let $E \subseteq [\alpha, \infty)$ be measurable. Put*

$$m = \int_E e^{-t} dt, \quad n = \int_E e^{-2t} dt.$$

Then

$$n \geq \frac{m^2}{2}.$$

Proof. Under the change of variables $w = e^{-t}$, the set E maps to a measurable set $A \subseteq (0, e^{-\alpha}]$, and

$$m = \int_A dw, \quad n = \int_A w dw.$$

Among subsets of $(0, e^{-\alpha}]$ of Lebesgue measure m , the integral of w is minimized by the interval $(0, m)$. Hence

$$n \geq \int_0^m w dw = \frac{m^2}{2}.$$

□

Lemma A.4 (Continuous plateau-set domination). *For every measurable set $E \subseteq [1/2, \infty)$,*

$$\int_E e^{-t} dt \geq \int_E \lambda(t) dt + 2 \iint_{E \times E} \lambda(u+v) du dv.$$

Proof. Write

$$E_0 := E \cap [1/2, 1), \quad E_1 := E \cap [1, \infty),$$

and define

$$m_i = \int_{E_i} e^{-t} dt, \quad n_i = \int_{E_i} e^{-2t} dt \quad (i = 0, 1).$$

Let $m = m_0 + m_1$ and $n = n_0 + n_1$. Since $u, v \geq 1/2$, we have $u + v \geq 1$, and therefore

$$\lambda(u+v) = Pe^{-(u+v)} - Be^{-2(u+v)}.$$

Also $\lambda(t) = 0$ for $t < 1$. Hence the difference between the left-hand side and the right-hand side is

$$G = m - Pm_1 + Bn_1 - 2Pm^2 + 2Bn^2.$$

By Lemma A.3, applied to E and to E_1 ,

$$n \geq \frac{m^2}{2}, \quad n_1 \geq \frac{m_1^2}{2}.$$

Thus

$$G \geq m - Pm_1 + \frac{B}{2}m_1^2 - 2Pm^2 + \frac{B}{2}m^4. \quad (9)$$

For fixed m , the right-hand side of (9) is decreasing in $m_1 \in [0, r^2]$, because

$$-P + Bm_1 \leq -P + Br^2 < 0.$$

Since $0 \leq m_1 \leq \min\{m, r^2\}$ and $0 \leq m \leq r$, the worst case is $m_1 = \min\{m, r^2\}$.

If $0 \leq m \leq r^2$, then (9) is bounded below by

$$L_1(m) := (1 - P)m + \left(\frac{B}{2} - 2P\right)m^2 + \frac{B}{2}m^4.$$

Since $B/2 - 2P < 0$ and $m \leq r^2$,

$$L_1(m) \geq m \left((1 - P) + \left(\frac{B}{2} - 2P\right)r^2 \right) \geq 0,$$

by the coefficient identity displayed in the remark after Lemma 5.2.

If $r^2 \leq m \leq r$, then (9) is bounded below by

$$L_2(m) := m - Pr^2 + \frac{B}{2}r^4 - 2Pm^2 + \frac{B}{2}m^4.$$

We have

$$L_2(r) = 0, \quad L_2(r^2) = (r - 1)(10r^4 + 4r^3 - 10r^2 - r + 2) > 0.$$

Moreover

$$L_2''(m) = -4P + 6Bm^2 \leq -4P + 6Br^2 = \frac{4(23r - 14)}{r^2} < 0$$

on $[r^2, r]$. Hence L_2 is concave with nonnegative endpoint values, and so $L_2(m) \geq 0$ throughout $[r^2, r]$. This proves $G \geq 0$. $\square$

Lemma A.5 (Continuous plateau functional domination). *For every nonnegative measurable F ,*

$$\int_{1/2}^{\infty} e^{-t} F(t) dt \geq \int_1^{\infty} \lambda(x) F(x) dx + 2 \int_1^{\infty} \lambda(x) \int_{1/2}^{x-1/2} \min\{F(y), F(x-y)\} dy dx.$$

Proof. For $s \geq 0$, let

$$E_s := \{t \geq 1/2 : F(t) > s\}.$$

Apply Lemma A.4 to E_s and integrate over $s \geq 0$. Tonelli's theorem gives

$$\int_0^{\infty} \int_{E_s} e^{-t} dt ds = \int_{1/2}^{\infty} e^{-t} F(t) dt,$$

$$\int_0^\infty \int_{E_s} \lambda(t) dt ds = \int_1^\infty \lambda(t) F(t) dt,$$

and

$$\int_0^\infty \iint_{E_s \times E_s} \lambda(u+v) du dv ds = \iint_{u,v \geq 1/2} \lambda(u+v) \min\{F(u), F(v)\} du dv.$$

Changing variables $x = u + v$ in the last double integral gives the displayed inequality. $\square$

A.3 Proof of the continuous theorem and sharpness

Proof of Theorem A.1. Assume $\int_0^\infty e^{-x} F(x) dx < \infty$, since otherwise there is nothing to prove. Multiply (7) by $\lambda(x)$ and integrate over $[1, \infty)$. By (8),

$$\begin{aligned} \gamma &= \int_1^\infty x \lambda(x) dx \\ &\leq \int_1^\infty \lambda(x) F(x) dx + 2 \int_1^\infty \lambda(x) \int_0^x \min\{F(y), F(x-y)\} dy dx. \end{aligned}$$

Split the inner integral into

$$[0, 1/2], \quad [1/2, x - 1/2], \quad [x - 1/2, x].$$

The middle contribution is exactly the second term in Lemma A.5. For the two edge contributions, use $\min\{F(y), F(x-y)\} \leq F(y)$ in the left edge and the change of variables $z = x - y$ in the right edge. Together with (8), the two edge contributions are bounded above by

$$4\Lambda \int_0^{1/2} F(y) dy = C \int_0^{1/2} F(y) dy.$$

By Lemma A.2, this is at most

$$\int_0^{1/2} e^{-y} F(y) dy.$$

By Lemma A.5, the λF term plus the middle contribution is at most

$$\int_{1/2}^\infty e^{-y} F(y) dy.$$

Therefore

$$\gamma \leq \int_0^\infty e^{-y} F(y) dy,$$

which proves the lower bound in Theorem A.1.

It remains to show sharpness. Let

$$F_*(x) = \min\{x, 1/2\}.$$

This function is nonnegative and subadditive. If $0 \leq x \leq 1/2$, then

$$F_*(x) + 2 \int_0^x \min\{F_*(y), F_*(x-y)\} dy = x + \frac{x^2}{2} \geq x.$$

If $1/2 \leq x \leq 1$, then

$$F_*(x) + 2 \int_0^x \min\{F_*(y), F_*(x-y)\} dy = \frac{1}{2} + \frac{x^2}{2} \geq x.$$

If $x \geq 1$, then

$$\int_0^x \min\{F_*(y), F_*(x-y)\} dy = \frac{x}{2} - \frac{1}{4},$$

and hence equality holds in (7). Finally,

$$\begin{aligned} \int_0^\infty e^{-x} F_*(x) dx &= \int_0^{1/2} x e^{-x} dx + \frac{1}{2} \int_{1/2}^\infty e^{-x} dx \\ &= 1 - e^{-1/2}. \end{aligned}$$

Thus the bound is sharp. □

References

- [1] Jannis Blauth, Meike Neuwöhner, Luise Puhmann, and Jens Vygen. Improved guarantees for the a priori TSP. *Mathematics of Operations Research*, 50(4):2909–2940, 2025. DOI: 10.1287/moor.2023.0322; arXiv:2309.10663.
- [2] Open Problems in Operations Research. Determine the exact master route ratio for a priori TSP with depot. Entry W4404566852_p0, accessed May 29, 2026. https://pranav-nuti.github.io/open-problems-in-or/problem.html?id=W4404566852_p0.